

IMPACT OF BANK INCOME DIVERSIFICATION ON BANK PERFORMANCE:

**Evidence from Licensed Commercial Banks in Sri
Lanka**



By

B.T.R.De Silva

MF/2021/6794

Department of Accountancy

Dissertation Presented to the Faculty of Management and Finance
of University of Ruhuna
in Partial Fulfillment of the Requirements for the Degree of
Bachelor of Business Administration
Honors in Accountancy

Table of Contents

Table of Contents	i
Abstract	iv
CHAPTER 1	1
INTRODUCTION	1
1.2 Background of the study	1
1.1.1 Local Context.....	1
1.1.2 International Context	2
1.2 Problem	2
1.3 Research Question	3
1.4 Research Objectives.....	4
1.4.1 General Objective	4
1.4.2 Specific Objectives	4
1.5 Significance and rational of the study.....	4
1.6 Summary and outline of the study	5
CHAPTER TWO	5
LITERATURE REVIEW.....	5
2.1 Chapter introduction	5
2.2 Concepts.....	6
2.2.1 Definition Of Concepts	7
2.3 Theories and Models.....	8
2.3.1 Modern Portfolio Theory (MPT)	8
2.3.2 Economies of Scope Theory	8
2.3.3 Agency Theory.....	9
2.4 Empirical review	9

2.4.1 Foreign Studies	9
2.4.2 Local Studies.....	10
2.5 Empirical gap	12
2.6 Chapter Summary	12
CHAPTER THREE	13
RESEARCH METHODOLOGY.....	13
3.1 Chapter Introduction	13
3.2 Sample & Sample Selection.....	14
3.2.1 Sample in Relation to Population	14
3.2.2 Sample Selection Technique and Justification.....	14
3.3. Data and Data Collection	16
3.4 Conceptual framework.....	16
3.5 Operationalization of Concepts and Variables	17
3.6 Method and data analysis.....	18
3.7 Chapter Summary	18
CHAPTER FOUR.....	19
DATA ANALYSIS AND DISCUSSION	19
4.1 Introduction.....	19
4.2 Descriptive Statistics.....	19
4.3 Normality Test.....	20
4.4 Analysis of Correlation	25
4.5 Reliability Analysis	26
4.6 Multicollinearity Test	27
4.7 Panel Unit Root Test	28
4.8 Panel Data Regression Analysis.....	29
4.8.1 ROA Model.....	30
4.8.2 ROE Model	32

4.10 Summery of findings.....	36
4.11 Chapter Summary	37
CHAPTER FIVE	38
CONCLUSIONS AND SUGGESTIONS.....	38
5.1 Chapter Introduction	38
5.2 Conclusions.....	38
5.3 Suggestions	39
5.4 Chapter Summary	41
References.....	42

Abstract

The main objective of this study is to examine the impact of income diversification on the financial performance of licensed commercial banks operating in Sri Lanka from the year 2019 to 2023. The main intention behind conducting this study is to evaluate the impact of income generated from various sources other than interest income on the financial performance of commercial banks, considering the Return on Assets (ROA) and Return on Equity (ROE) as indicators of financial performance. This study adopted the Fixed Effects (FE) and Random Effects (RE) regression models using a panel data set of ten commercial banks operating in Sri Lanka to examine the impact of income diversification on the financial performance of commercial banks operating in Sri Lanka. This study also adopted a few control variables, namely the size of the commercial banks, represented by total assets, to evaluate their influence on the financial performance of commercial banks operating in Sri Lanka. The results of this study revealed that income diversification has a negative but statistically insignificant influence on the financial performance of commercial banks operating in Sri Lanka, considering the ROA and ROE indicators. This study also showed that financial leverage had a positive marginal significance on ROE, which showed that financial leverage had a marginal impact on the enhancement of financial performance. This study showed that bank size was not significant in the enhancement of financial performance, which showed that bank size had no significant impact on the enhancement of financial performance. This study showed that income diversification had a potential impact on the enhancement of financial stability, but it had a marginal impact on the enhancement of financial performance, which showed the need to manage banks strategically to enhance financial performance. This study also provided significant implications to bank managers, policymakers, and regulators who are interested in exploring income strategies to enhance financial performance in emerging markets, including Sri Lanka. This study also showed that future researchers need to incorporate foreign banks, financial indicators, and time periods while exploring income diversification and financial performance dynamics in the Sri Lankan banking industry.

Keywords: Income Diversification, Bank Performance, ROA, ROE, Financial Leverage

CHAPTER 1

INTRODUCTION

1.2 Background of the study

Over the last two decades, commercial banks have drawn considerable attention regarding their capacity to promote efficiency in the financial system, economic growth, and financial stability. The banking business model has traditionally been dominated by interest-based activities, primarily deposit mobilization and loan disbursement. However, under the influence of technological innovations, regulatory reforms, financial liberalization, and increased competition, banks have increasingly diversified their income structures towards non-interest-based activities such as trading, brokerage, fee-based services, asset management, and investment banking. An earlier concept was that this tendency for diversification could be considered a method of decreasing volatility in income and increasing profitability to reduce risks for better long-term performance (Jayasena & Amarasinghe, 2018).

1.1.1 Local Context

Income diversification has become a dynamic strategic tool that licensed commercial banks use in their response to pressures for macroeconomic instability, increasing non-performing loans, eroding interest margins, and fierce competition from non-bank financial institutions. For instance, Jayasena and Amarasinghe (2018) reported the significant positive influence of income diversification on bank performance across ten licensed commercial banks during 2006-2016 and found that diversified sources of income improved both ROA and ROE. Similarly, Pisedtasalasai and Edirisuriya (2020) reported the profitability and long-term strength of Sri Lankan banks with more diversified income structures.

However, some contradictory findings can also be seen in the Sri Lankan environment. For instance, Menike (2022) found that portfolio diversification had negatively influenced profitability, especially ROE for ten banks listed from 2012 to 2022. Further,

Kurincheedaran (2015) emphasized that sectoral loan diversification will drive the banks to poorer performance, with especial reference to state-owned banks, and therefore inferred that excessive diversification will weaken focus and operational efficiency.

These contradictory results thus suggest that, up to date, the Sri Lankan banking industry has not reached a unanimous conclusion on whether diversification benefits or hinders bank performance, making this topic extremely relevant and timely for research.

1.1.2 International Context

Not surprisingly, the overall implications of income diversification for bank performance remain mixed globally. For example, the studies conducted for emerging economies like Vietnam document that diversification has a negative effect on liquidity creation and profitability; hence, banks should be more focused on traditional lending activities. In contrast, evidence from developed economies suggests that diversified financial institutions are likely to be more resilient and achieve higher long-term profitability because of multiple sources of revenue that decrease dependence on interest-based income.

As was powerfully stated in a range of global financial literature: It is in this context that Hoang et al. (2020) assert, "In emerging markets, diversification may increase financial fragility when performed without adequate risk-management capabilities". This underlines a global concern that diversification, theoretically desirable, may become problematic for countries with weaker regulatory frameworks, limited technological capacity, and volatile macroeconomic conditions-common characteristics of transition economies like Sri Lanka. As such, the Sri Lankan case provides a strong contextual framework through which diversification strategies influence bank performance under local market, regulatory, and risk conditions.

1.2 Problem

Even though there is an emerging interest, both globally and locally, in the diversification of income in banking strategy, the empirical results for bank

performance are inconsistent. Some studies in Sri Lanka report a positive effect, whereas others reveal the negative or insignificant impacts (Jayasena & Amarasinghe, 2018; Pisedtasalasai & Edirisuriya, 2020). The inconsistency would add confusion to those policymakers, banking managers, and regulators relying on evidence-based strategies for the sustainability of banks' profitability and risk management practices. Moreover, the heightening financial vulnerability in the post-crisis economic environment of Sri Lanka demands the urgency of assessing the effectiveness of the diversification strategies (Menike, 2022; Kurincheedaran, 2015).

The challenge therefore emerges from a critical knowledge gap: *Does income diversification genuinely enhance bank performance in the Sri Lankan context, or does it expose banks to greater instability and reduced profitability?* According to CBSL reports, the financial sector has been increasingly pressured due to macroeconomic imbalances, market volatility, and reduced credit flows, making income diversification both a necessity and a potential risk factor. This constitutes a strong statement of national relevance: Sri Lanka's banking sector faces an international-level competitiveness and stability challenge, reinforcing the urgency to understand how diversification strategies influence performance.

Hence, this study addresses the research problem through empirically analysing how income diversification has influenced the performance of licensed commercial banks in Sri Lanka.

1.3 Research Question

- What is the impact of **income diversification (non-interest income)** on the financial performance of commercial banks?
- How does **leverage (total liabilities)** affect the profitability of commercial banks?
- How does income diversification influence **Financial performance** of commercial banks?

1.4 Research Objectives

1.4.1 General Objective

To examine the impact of income diversification, leverage on the financial performance of commercial banks.

1.4.2 Specific Objectives

- To analyze the effect of non-interest income on bank performance.
- To examine the relationship between total liabilities and bank profitability.
- To evaluate the influence of income diversification on Financial performance.

1.5 Significance and rational of the study

The contribution of this study can be viewed in threefold importance for academic research, policy making, and practical banking operations. Theoretically, the study contributes to an improved understanding of the diversification-performance nexus in emerging markets. Given that prior studies in Sri Lanka have generated inconsistent results, this current study fills an important gap in the literature by presenting updated and context-specific evidence.

The findings also carry practical importance for banking managers, as they will need to make strategic choices regarding the balancing of traditional and non-traditional incomes. In a context of depressing interest margins, increased competition, and economic instability, knowing if diversification supports profitability or not will help banks achieve long-term sustainability. Moreover, the current research performance gap, as identified in the literature, exists because there is a mismatch between diversification practices and real financial outcomes, thus allowing banks to refine their risk management strategies and further enhance resource allocation. The evidence from this study will be useful for policymakers and regulators in deciding whether the regulatory frameworks should encourage or restrain diversification activities, particularly in view of their implications for systemic risk. In summary, the suggestions of the research study support stable and efficient banking operations and contribute to the stability of the financial system.

1.6 Summary and outline of the study

This research is organized into five integrated chapters that collectively navigate the investigation. Chapter One introduces the background of the study, states the research problem, and provides the research questions and objectives. It also provides the significance of the research on income diversification within the Sri Lankan banking context.

In summary, the introductory chapter builds a case for the research and provides reasons why the topic is of importance to academics, practitioners, and policy analysts. Chapter Two gives an in-depth literature review concerning income diversification and bank performance. It discusses the underlying theories, conceptual developments, and empirical findings from both local and international studies. This chapter synthesizes prior research and thus provides a theoretical and empirical framework supportive of developing the conceptual model of the study. Chapter Three outlines the methodology, where one can find the research design, data selection, variables, and the analytical techniques employed to analyse the effect of income diversification on commercial bank performance. Chapter Four presents the results of the empirical analysis and gives interpretations concerning the literature reviewed earlier. It also reflects on whether or not the findings agree with or differ from previous studies. Lastly, Chapter Five summarizes key findings, gives practical recommendations, and discusses implications for future research and policy development. Together, these ensure coherence in a systematic exploration of the research topic.

CHAPTER TWO

LITERATURE REVIEW

2.1 Chapter introduction

Income diversification has proven to be a notable tactical adoption made by commercial banks to improve their performance and make their finances more stable in the current highly competitive and instable environment. Conventional commercial banks used to focus on interest income earned from their loan business; but in this era of globalization

and technological advancement, banks are forced to pursue non-interest income sources in order to compensate for their growing loan business expenses and adapt to the high degree of competitiveness in the industry. Within the Sri Lankan banking industry, commercial banks operating with a licensed status play an important role in the development of the national economy.

This chapter presents an investigation into existing literature concerning income diversification, bank performance, with special reference to licensed commercial banks in Sri Lanka. The basis or origin of this research shall flow from this chapter, where basic concepts shall be explained, together with comparisons concerning major constructs. Models pertinent to this research shall then follow.

2.2 Concepts

The core theme of the research is income diversification and its impact on bank performance. The variable of income diversification captures the dynamic nature of income sources after which instead of focusing on interests-only sources, various other sources could also be considered. The overall bank performance was measured on the basis of accounting metrics like Return on Assets (ROA), Return on Equity (ROE), whereas bank risk acts like a moderating variable for the outcomes of diversification. Size and bank ownership were controlled for to check the assumption of this hypothesis for both size and ownership of the bank.

Findings from empirical research carried out in Sri Lanka are quite mixed in terms of proving the advantages of income diversification. For instance, Jaysena and Amarasinghe (2018) identified that income diversification has a positive association with bank performance in licensed commercial banks. Likewise, Menike (2024) identified that income diversification has a positive association with the profitability of Sri Lankan private commercial banks. However, income diversification fails to affect bank performance after accounting for ownership effects as identified by Ekanayake and Wanniarachchige (2023).

2.2.1 Definition of Concepts

"Income diversification refers to the degree to which banks diversify their revenue streams beyond interest income to other types of non-interest incomes" (Stiroh & Rumble, 2006).

"In defining diversification for the banking industry, particular attention must be paid to the nature of bank income." Roger Bolton argues that banks that have diversified their activities will have different types of income, which in turn require changes in banking management. He defines banking diversification in "Revenue diversification is the share of non-interest income in total operating income of a bank" (DeYoung & Roland, 2001).

Although both define income diversification, the second uses identifiable income parts that can be quantified, which can be subjected to empirical testing. These two definitions overall specify that income diversification is not solely an extension but a quantifiable strategy.

"Bank performance measures a bank's capacity to efficiently create profits out of its assets and equity" (Athanasoglou, Brissimis, & Delis, 2008).

"Return on Assets and Return on Equity are widely used indicators to measure managerial efficiency and profitability of banks" (Goddard, Molyneux, & Wilson, 2004).

Both definitions stress profitability and efficiency. ROA focuses on asset management efficiency, and ROE is related to earnings on capital. Both ratios give a complete insight into banking efficiency.

"Bank risk is defined as the variability of returns and the risk of financial distress that can emerge from banking activities" (Saunders & Cornett, 2019).

"Financial stability is defined as the ability of banks to withstand shocks without impairing financial intermediation" (IMF, 2017).

Risk is more concerned with uncertainty and variability, while financial stability is an indication of its resilience against adverse conditions. Income diversification is perceived to minimize risks through sharing but may accentuate variability associated with non-traditional activities.

"Bank size can be measured by the natural logarithm of total assets, which represents economies of scale and market power," according to Berger & Humphrey (1997).

The larger banks are believed to have more benefits from the diversification, contrary to the small banks. This is attributed to the availability of resources and the capabilities to operate..

“Ownership structure captures differences among banks depending on state versus private ownership,” La Porta et al. explain (2002).

In those cases, public banks can be driven by objectives such as stabilization and growth, whereas the goals of private banks could be more profit-focused.

2.3 Theories and Models

2.3.1 Modern Portfolio Theory (MPT)

The Modern Portfolio Theory, which emerged as a result of the work of Markowitz (1952), states that rather than increasing risks, diversification has the effect of reducing the overall risks that are associated with the imperfectly correlated returns of assets.

Based on this principle, the earning diversification strategy helps to mitigate risks that are associated with banking as this strategy ensures that the banks are able

Within the framework of licensed commercial banks in Sri Lanka, diversification of earnings could reduce potential vulnerabilities arising from credit risk and economic slumps. Kurincheedaran (2015) and Pisedtasalasai and Edirisuriya (2020) proposed similar views through their findings that diversified institutions outperform others after adjusting risk-adjusted measures by using MPT.

2.3.2 Economies of Scope Theory

According to the economies of scope theory, a company can lower its costs through the joint production of two or more different products rather than focusing on a single product (Panzar & Willig, 1981). For a bank, providing a range of financial services helps in achieving a common cost.

The commercial banks in Sri Lanka, offering lending services, payments, wealth management, and trade finance solutions, may gain from the potential synergy benefits. Small banks may not gain from these benefits owing to size limitations, thereby highlighting the significance of bank size as a moderator in this research.

2.3.3 Agency Theory

In Agency theory examines conflicts of interests between managers and shareholders as identified by Jensen & Meckling in 1976. Managers become interested in diversification for increasing their own power, mitigating employment risk, or providing a rationale for expansion, even when it doesn't increase shareholder value. in banking, over-diversification of earnings, particularly in non-interest earning activities, could be risk-increasing or performance-reducing.

This explanation encompasses contradictory empirical outcome findings in the context of Sri Lanka, with some research claiming positive outcomes (Menike, 2024) while others discovering insignificant or negative correlations (Ekanayake & Wanniarachchige, 2023).

2.4 Empirical review

Income diversification and its impact on bank performance remains an area that has received mixed findings from empirical research studies due to differences in banking systems and structures among other factors. The relationship between income diversification and bank profitability, risk, liquidity creation ability, and bank financial stability has been tested by previous researchers using accounting-based performance measures such as ROA and ROE. This section reviews empirical studies by foreign and local researchers aligned with the objectives of this study.

2.4.1 Foreign Studies

There is a significant international literature that examines the effect of income diversification on bank performance and risk. The early empirical studies by Stiroh (2004) showed that greater dependence on non-interest income increased earnings

volatility, which in turn lowers risk-adjusted returns in U.S. banks. Then, Berger et al. (2010) showed that diversification into non-traditional activities can lead to risk exposure, despite its possible benefits.

Conversely, some studies have found positive results of diversification. Li and Zhang found that income diversification has a positive effect on banking performance because of its stabilizing influence on income and reduced sensitivity to net interest income. The study supports the portfolio theory of diversification that a diversification of income bases is linked to higher profitability.

However, evidence from economies in emergence is more mixed. Sharma & Anand (2018) investigated banks operating in BRICS countries and found that income diversification had a negative impact on bank performance, as ROA & ROE. These researchers supported that lack of quality infrastructure and managerial knowledge in economies in emergence contour little effect on income diversification.

One of the key findings is the work done by Hoanga et al. (2020) on the Vietnamese commercial banks. Using panel data OLS and GMM, the authors of the study concluded that income diversification negatively influenced liquidity creation and financial performance. According to the findings, the authors recommend that Vietnamese banks stick to the core practice of lending, as the volatile nature of non-interest income decreased efficiency. This is unusual, as portfolio theory suggests the reverse.

Equally, the impact of income diversification on ROA and ROE in the Indian banking sector was studied by Vidyarthi (2020). Its outcome indicated that the diversification of income had no significant or negative effect on ROA and ROE in times of economic distress.

In general, foreign empirical studies show that income diversification does not have any kind of optimal effect on bank performance, as positive, negative, and insignificant relationships have been found.

2.4.2 Local Studies

The empirical research undertaken in Sri Lanka is significant in understanding the relationship between income diversification and performance in licensed commercial banks. Jayasena and Amarasinghe in 2018 engaged in panel data analysis among

licensed commercial banks in Sri Lanka for a period of 2006 to 2016. The research brought forth significant outcomes, affirming that income diversification has a significant and positive influence on both ROA and ROE in licensed commercial banks in Sri Lanka.

Additionally, more contemporary proof, as identified in Menike (2024), further supports this result. The research confirmed the existence of a positive relationship between revenue diversification and the performance of Sri Lankan Private Commercial banks, establishing that revenue diversification helps to result in better profits.

However, not all studies at the local level indicate that these results are correct. A study conducted by Pisedtasalasai and Edirisuriya in 2020, which investigated the impact of income diversification, taking into consideration the impact of ownership structure, revealed that income diversification leads to improved performance in privately-owned banks but has a less significant impact on state-owned banks.

Ekanayake and Wanniarachchige (2023) similarly found that income diversification does not significantly impact banking performance after controlling for ownership effects. Such a result contradicts previous results and implies that ownership structures and policies affect income diversification.

Research that investigates risk and diversification in portfolios also appears mixed. Kurincheedaran (2015) established that diversification by sectors increases bank performance, whereas Kumanayake & Gunarathne (2019) argued that diversification by loans influences bank profitability in a positive manner but could also elevate risk levels if not handled effectively.

comparatively, the positive findings related to the benefits of income diversification are more visible in the local studies, especially with regard to ROA and ROE, as compared to the foreign studies. This could be due to the fact that the banks of Sri Lanka have been quite conservative with regard to diversification, among other reasons.

When compared with other studies conducted globally, including emerging markets, there is a contrast in findings. For instance, studies in other countries tend to point towards the negative or insignificant impact of diversification, whereas in Sri Lankan

studies, it is noticed that a positive relationship generally exists between income diversification and banking performance.

Despite some disparities, most of the empirical research within the Sri Lankan context tends to support the view that income diversification is a positive determinant of bank profitability.

2.5 Empirical gap

Although several research papers cover the relationship of income diversification to bank performance, several research gaps are notable. Firstly, most research papers concerning Sri Lanka utilize comparatively older research datasets, whereas recent structural and regulatory shifts in the sector are largely unreflected. Secondly, insufficient consideration has been placed on the cumulative relationship concerning profitability, risk, as well as stability in one framework. Thirdly, the research concerning whether the relationship concerning income diversification operates in a size and ownership-stratified manner to banks remains inconclusive.

Furthermore, the inconclusive results of both local and overseas research point towards the need for new and extensive research on the topic of licensed commercial banks in Sri Lanka. Closing these research gaps forms the basis of the current research.

2.6 Chapter Summary

The current chapter has covered the theoretical and existing literature concerning income diversification and performance of banks. The existing literature has been able to demonstrate that many studies have reported divergent results, where foreign studies have concentrated more on negative or insignificant impacts, and Sri Lankan studies have supported a positive outcome concerning income diversification and performance of banks. The chapter has noticed important gaps and thus illustrated the importance of exploring this topic further. The approaching chapter introduces the methodology used to explore these relationships within the Sri Lankan banking sector.

CHAPTER THREE

RESEARCH METHODOLOGY

3.1 Chapter Introduction

This chapter presents the research methodology employed to explore the impact of income diversification strategies on the performance of licensed commercial banks in Sri Lanka. The proposed research methodology encompasses robust strategies to ensure research reliability, validity, and relevance. The current chapter includes subtopics that discuss the sample, procedures in determining sample selection, procedures in which data can be accumulated, framework, definition, measurement, procedures employed to perform analysis, and data analysis procedures. The proposals and procedures under consideration are purely quantified since financial analysis is performed through

financial statements, which are extracted from financial accounts that have been audited within a stipulated time period.

3.2 Sample & Sample Selection

3.2.1 Sample in Relation to Population

The population in this research study includes all licensed commercial banks (LCBs) in Sri Lanka operating under the regulation of the Central Bank of Sri Lanka. As defined by the classification by the Central Bank of Sri Lanka, licensed commercial banks comprise national public banks, national private commercial banks, and foreign banks. It is not possible to study the whole population, and so the study selects a representative sample.

The sample comprises licensed commercial banks, and collectively, they account for more than 50 percent of the total assets of the banking industry in Sri Lanka. This is a screening criterion that ensures the selected banks are a good representation of the overall banking industry and that the industry is diversified in terms of income and behavior. The selection of banks with relatively larger asset bases will enhance the overall generalizability of the findings of the study.

3.2.2 Sample Selection Technique and Justification

The approach adopted in order to choose the sampling banks is through purposive sampling. The approach adopted in this research falls under the type of non-probability sampling approach. The approach adopted fits well given that the research demands to see the financial report making similar disclosures. Additionally, the adopted approach demands that the financial institution should be operational during the entire period of the research.

Table 3.1: Sample of the Study

No.	Name of the Bank	Type of Bank	Study Period

1	Bank of Ceylon (BOC)	Licensed Bank	Commercial	2019–2023
2	Commercial Bank of Ceylon (CBC)	Licensed Bank	Commercial	2019–2023
3	Hatton National Bank (HNB)	Licensed Bank	Commercial	2019–2023
4	Sampath Bank PLC	Licensed Bank	Commercial	2019–2023
5	Nations Trust Bank PLC	Licensed Bank	Commercial	2019–2023
6	Seylan Bank PLC	Licensed Bank	Commercial	2019–2023
7	DFCC Bank PLC	Licensed Bank	Commercial	2019–2023
8	National Development Bank PLC (NDB)	Licensed Bank	Commercial	2019–2023
9	Peoples bank	Licensed Bank	Commercial	2019–2023
10	Amana Bank PLC	Licensed Bank	Commercial	2019–2023

Both state-owned and private domestic commercial banks are taken into account for the research study. The state-owned banks are chosen based on their dominant role in financial transaction, development, and importance to the banking sector of Sri Lanka. Private domestic banks are also taken into account based on their for-profit approach to making money and increasing their involvement with income diversification schemes. Both groups can be included to help investigate differences between the two groups with regard to income diversification.

Foreign commercial banks are excluded from the sample for a set of reasons. First, the character of the operation of the foreign banks in Sri Lanka is that they function through branches; therefore, they are not incorporated in Sri Lanka. Hence, they are not granted further financial information. Second, the source of their income, as well as their strategies, are dominated by the global character of business; therefore, they are not comparable to local banks. Third, their share in the total assets as well as their advances in Sri Lanka are relatively smaller; therefore, they are not appropriate for analysis. The exclusion of the foreign commercial banks from the analysis for a similar set of reasons supports existing literature on banking in Sri Lanka (Jayasena & Amarasinghe, 2018; Pisedtasalasai & Edirisuriya, 2020).

3.3. Data and Data Collection

The study utilizes secondary data. The financial information has been obtained from the published annual reports of the selected licensed commercial banks. The other information has been obtained from the published annual reports and financial stability reports of the Central Bank of Sri Lanka. The utilization of secondary data is justified since it provides audited, reliable, and truthful information.

The observation period is several years, depending upon the levels of income diversification. Secondary panel data in previous empirical work on bank performance and diversification data has been extensively used because of appropriateness in terms of trends as well as cross-sectional information (Berger et al., 2010; Vidyarthi, 2020).

3.4 Conceptual framework

The conceptual framework for this paper has been formed on the basis of theoretical and empirical literature. The income diversification that is measured in terms of total income from various sources divided by total operating income is the independent variable. Bank performance is the dependent variable and is formed in terms of Return on Assets (ROA) and Return on Equity (ROE).

Conceptual framework

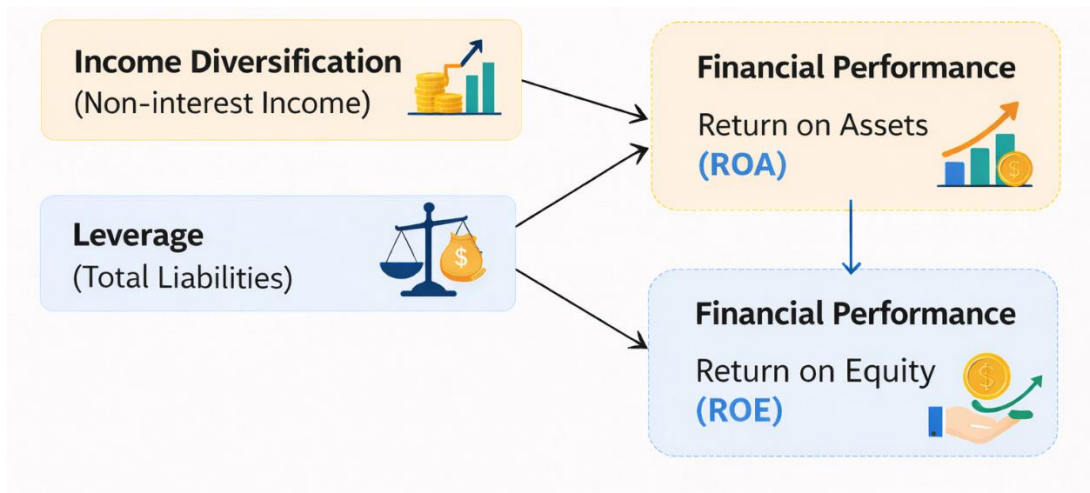


Figure 3.1: Conceptual framework

3.5 Operationalization of Concepts and Variables

Concept	Variable	Measurement	Source
Income Diversification	Diversification Index	Non-interest income / Total operating income	Stiroh (2004); Jayasena & Amarasinghe (2018)
Bank Performance	ROA	Net profit / Total assets	Athanasoglou et al. (2008)
Bank Performance	ROE	Net profit / Shareholders' equity	Goddard et al. (2004)
Financial Leverage	Leverage	Total liabilities / Total assets	Menike (2024)

3.6 Method and data analysis

The design employed in this research is quantitative in nature, which utilizes techniques related to the analysis of regression equations involving panel data. The employment of descriptive statistical analysis will help in grasping an idea concerning data, with an interpretation of the distribution of data. The implementation of correlation analysis will help in analyzing this research to determine the kind of variables. There is an aspect related to possible multicollinearity that can be identified by analyzing correlation.

The panel regression equations, fixed-effects model equations, and equations for the random-effects models are employed for estimating the impact of income diversification on bank performance. The Hausman test constitutes the basis for determining the method of estimation.

$$ROA_it = \beta_0 + \beta_1 ID_it + \beta_3 LEV_it$$

$$ROE_it = \beta_0 + \beta_1 ID_it + \beta_2 SIZE_it + \beta_3 LEV_it$$

ROA_it = Return on Assets for bank

ROE_it = Return on Equity for bank

ID_it = Income diversification \

LEV_it = Financial Leverage

The findings of significance tests are validated against normal values, and diagnosis tests are used to verify model validity.

3.7 Chapter Summary

This chapter has described the research methodology adopted in the research. This includes data collection by sampling, concept framework, operational definitions of the concepts, as well as analysis methods. The aforementioned research methodology has also been pertinent in the preceding empirical literature in banking, as already stated. This will be followed by the next chapter, which will present empirical results.

CHAPTER FOUR

DATA ANALYSIS AND DISCUSSION

4.1 Introduction

The present chapter will elaborate on the empirical analysis of the study, with special focus on the impact of income diversification on the performance of banks in the context of licensed commercial banks in Sri Lanka. In this regard, the study will use the panel data obtained from ten banks over the period ranging from 2019 to 2023, resulting in fifty data points. In this regard, the study will use the Stata software to conduct the analysis. In this particular chapter, the purpose is to develop a rigorous understanding of the impact of income diversification, as measured by the proportion of non-interest income to total operating income, on the overall performance of the bank, as indicated by the Return on Assets (ROA) and the Return on Equity (ROE). In this regard, the study will also use other variables such as financial leverage and the bank's size to develop a comprehensive understanding of the overall performance of the bank. In this regard, the study will use an analytical strategy that will begin with the distribution of the data, followed by the correlation and normality tests, and then the regression model, which will consist of the panel unit root test, the multicollinearity test, and the fixed effects and the random effects regression model, with the help of the Hausman test to select the most suitable model. In this regard, this particular chapter will help in developing a descriptive and inferential understanding of the data to support the research hypotheses in the context of the theoretical and conceptual understanding of the performance of the bank.

4.2 Descriptive Statistics

The above descriptive statistics give an initial overview of the distribution of the major variables of interest. Table 4.1 presents the mean, standard deviation, minimum, maximum, skewness, and kurtosis of ROA, ROE, income diversification, and leverage for the selected banks. The mean ROA is 0.969%, implying that the banks in Sri Lanka, on average, are earning 0.97% of their total assets as profit. The standard deviation of 0.423 suggests that the banks in Sri Lanka are moderately efficient in terms of their

profitability. In contrast, the mean ROE is 11.079%, implying that the banks in Sri Lanka are earning an average return of 11% to their shareholders. The highest return to the shareholders is 21.3%, implying that the banks in Sri Lanka are profitable to their shareholders. The income diversification ratio has an average of 0.187, implying that nearly 18.7% of the total income generated by the banks in Sri Lanka is not related to their interest income. This suggests that the banks in Sri Lanka are mainly dependent on their income generated from lending activities. The mean value of 0.867 for the total liabilities to total assets ratio suggests that the banks in Sri Lanka are financing their operations mainly through liabilities. The negative skewness value of -1.57 suggests that the banks in Sri Lanka are operating with high leverage. The high kurtosis value of 3.824 suggests that the banks in Sri Lanka are maintaining extreme values of leverage. Collectively, the descriptive statistics will provide an individual with an understanding of the characteristics of the sample population. The descriptive statistics provide the foundation for the other statistical tests.

Table 4.1 Descriptive Statistics of Variables

Variable	Mean	Std. Dev	Min	Max	Skewness	Kurtosis	Observations
ROA	0.969	0.423	0.25	2.60	1.204	6.094	50
ROE	11.079	4.562	3.50	21.30	0.485	2.783	50
Income Diversification	0.187	0.110	0.034	0.395	0.338	1.658	50
Leverage	0.867	0.109	0.593	0.947	-1.570	3.824	50

4.3 Normality Test

Normality of the variables is one of the aspects that should always be tested in the analysis of the panel data in order to ensure the reliability of the inferences obtained in the tests. Normality of the variables was tested using the skewness and kurtosis methods. The test was done using the following table. ROA has a skewness of 1.204, which is slightly positive, while the kurtosis is 6.094, which is quite high. This implies that the distribution of the ROA is not entirely normal since the majority of the banks are concentrated around the mean, while the remaining banks are either highly

profitable or less profitable. On the other hand, ROE is normally distributed since the skewness of 0.485 and the kurtosis of 2.783 are almost normal. This implies that the returns to the shareholders of the banks are almost normally distributed. Income diversification is slightly positively skewed with a skewness of 0.338 and a kurtosis of 1.658, which are almost normal. This implies that the income diversification is almost normally distributed across the banks in the country. This implies that the banks are not highly diversified in terms of income. On the other hand, the leverage is negatively skewed with a skewness of -1.57, which is quite high, while the kurtosis of 3.824 is slightly high. This implies that the distribution of the leverage is not entirely normal since the majority of the banks are highly leveraged, while the remaining banks are less leveraged.

Table 4.2 Normality Test Results

Variable	Skewness	Kurtosis	Normality Interpretation
ROA	1.204	6.094	Moderately skewed, leptokurtic
ROE	0.485	2.783	Approximately normal
Income Diversification	0.338	1.658	Slightly skewed
Leverage	-1.570	3.824	Negatively skewed

The above results indicate that the majority of the variables are normally distributed, except for ROA and leverage. Even though the results are slightly not normal, the reliability of the regression models is not affected, as the panel data models are robust to non-normality.

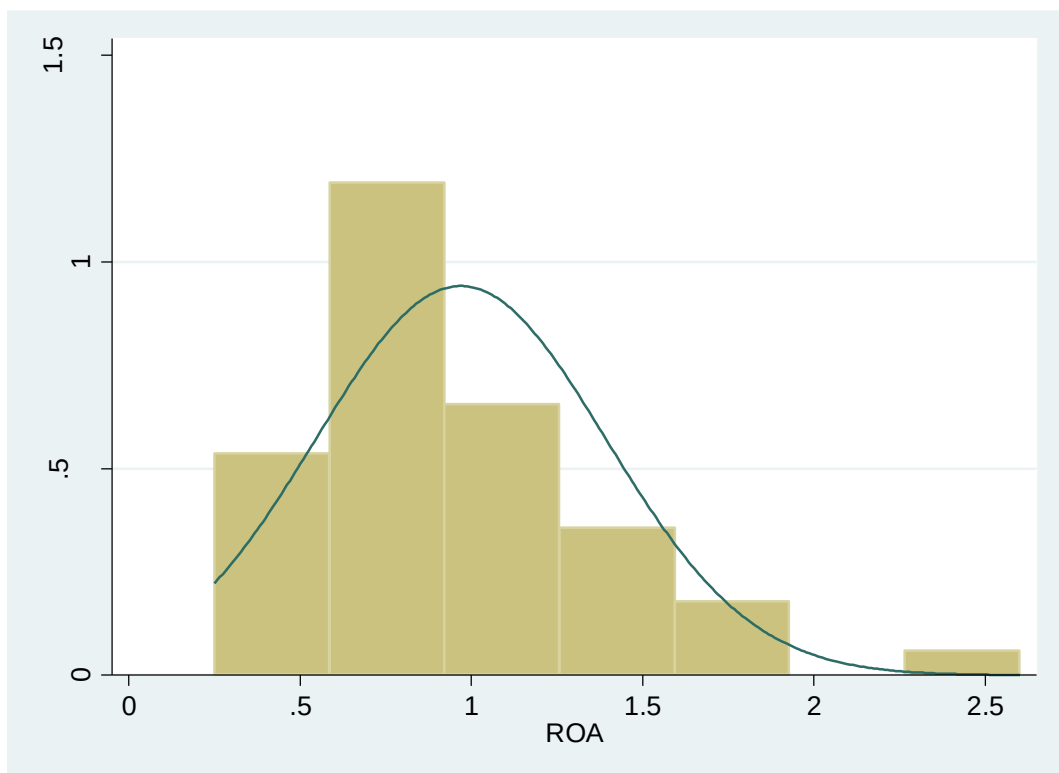


Figure 4.1: Return on Assets

This graph shows that ROA is slightly positively skewed and highly peaked. Most banks are clustered around the average level of performance, while only a few banks have extremely high or very low profitability.

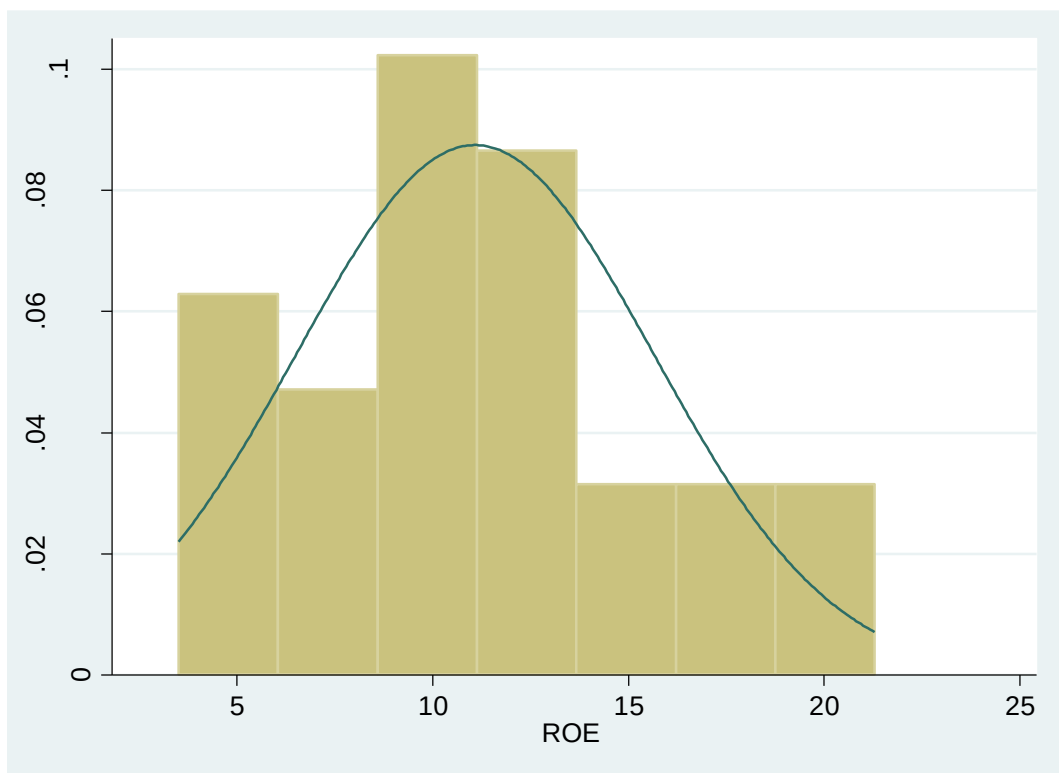


Figure 4.2: Return on Equity

This graph shows that ROE is approximately normally distributed with a slight positive skew. Most banks have similar levels of shareholder returns clustered around the average, with only minor variations and no extreme values.

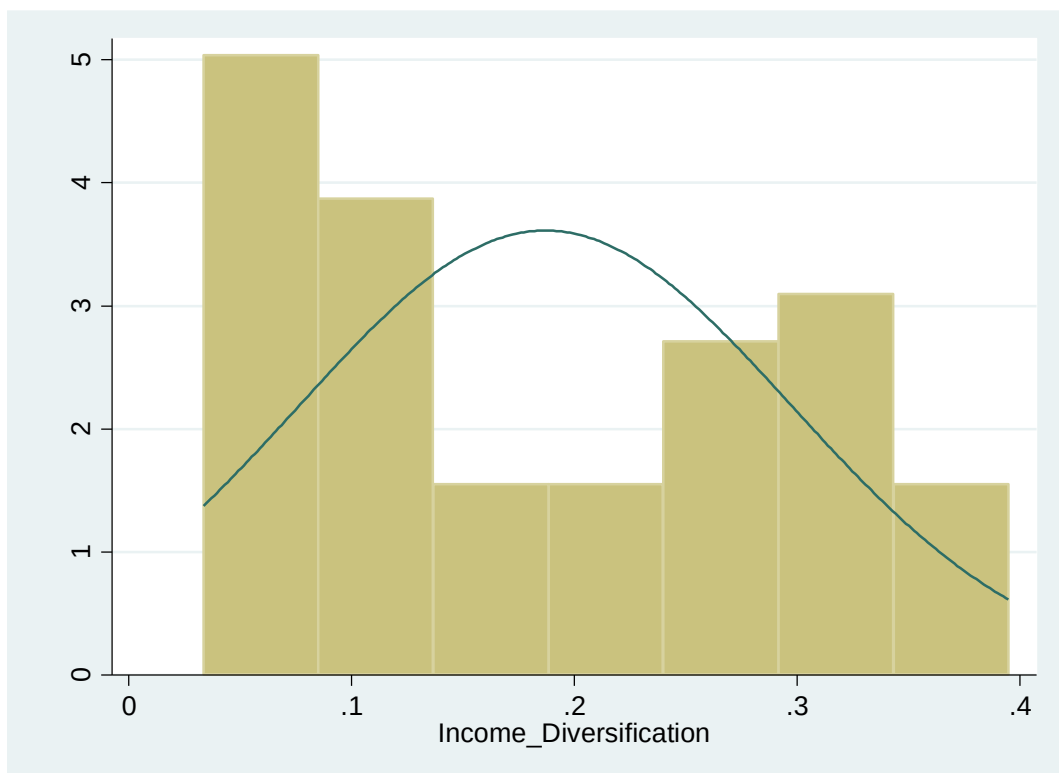


Figure 4.3: Income Diversification

This graph shows that income diversification is approximately normally distributed with a slight positive skew. Most banks have similar levels of income diversification, with no significant differences, and overall diversification remains relatively moderate

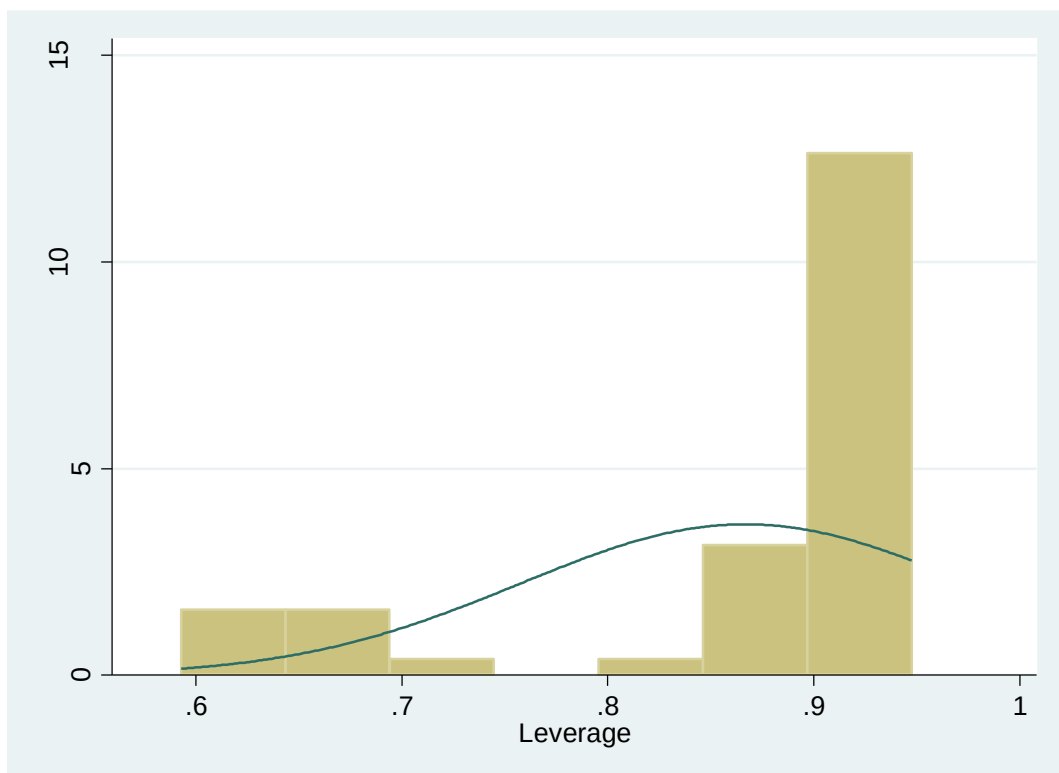


Figure 4.4: Leverage

This graph shows that most banks rely heavily on debt financing, resulting in higher leverage values. Only a small number of banks maintain lower leverage levels, which creates the negative skewness. The distribution is also slightly peaked, indicating that many banks have similar high leverage ratios.

4.4 Analysis of Correlation

Table 4.3 Correlation Matrix

Variable	ROA	ROE	Income Diversification	Leverage	Total Assets
ROA	1.000				
ROE	0.8157	1.000			
Income Diversification	0.0665	0.0091	1.000		
Leverage	0.2729	0.5316	0.3875	1.000	
Total Assets					1.000

Total Assets	0.0811	0.4666	-0.3758	0.5507	1.000
---------------------	--------	--------	---------	--------	-------

Correlation analysis was conducted to measure the degree and type of association between variables. Table 4.3 below shows the results of the correlation analysis. The correlation coefficient between ROA and ROE is strong and positive at 0.8157. This indicates that banks that have high returns on assets also have high returns on equity, which they pay to their shareholders. Income diversification is shown to have very low correlation with ROA at 0.0665 and low correlation with ROE at 0.0091. This indicates that income diversification may not have an impact on the profitability of the banks, at least for the banks in Sri Lanka. Leverage is shown to have moderate positive correlation with ROE at 0.5316. This indicates that banks that have high levels of debt relative to equity also have high returns on equity, which they pay to their shareholders, as theoretically expected. Leverage is also shown to have moderate correlation with total assets at 0.5507. This indicates that banks that have high levels of assets also have high levels of debt relative to equity, as if the high asset size enables the bank to afford high levels of debt. Income diversification is shown to have negative correlation with total assets at -0.3758. This indicates that banks that have high levels of assets may have low levels of income diversification, as if the high asset size enables the bank to afford high rates on loans, which results in high non-funded income for the bank.

4.5 Reliability Analysis

Table 4.4 Reliability Test (Cronbach's Alpha)

Items	Cronbach Alpha
ROA, ROE, Income Diversification, Leverage	0.201

Although the reliability of the financial ratios is not usually tested, the outcome of the test for internal reliability using the Cronbach's Alpha test was also done to test the internal reliability of the main variables (ROA, ROE, income diversification, and leverage). The outcome of the test showed that the value of the test, Cronbach's Alpha, was 0.201 (Table 4.4). The low value of the test is understandable because the financial ratios are independent measures of different aspects of the performance and operation

of the banking industry. The ROA and ROE ratios measure the profitability of the firm from different perspectives, income diversification measures the composition of income, and leverage measures the financial structure of the firm. The low value of the test, therefore, is not the outcome of the test showing the validity of the data collected from the financial statements of the firms, but rather the outcome of the test is confirming the fact that the data collected from the financial statements of the firms is reliable and that each variable is a measure of different aspects of the operation of the banking firms. The use of multiple variables is an added advantage because the robustness of the test is enhanced because of the multi-dimensional nature of the test of the factors influencing the performance of the banking firms.

4.6 Multicollinearity Test

Table 4.5 Multicollinearity Test (VIF)

Variable	VIF	1/VIF
Leverage	3.50	0.285
Total Assets	3.47	0.288
Income Diversification	2.84	0.352
Mean VIF	3.27	

Multicollinearity is a term used in regression analysis that refers to the correlation between two or more independent variables. Due to the presence of multicollinearity, the regression coefficient cannot be established. As a result, the standard error increases. In this study, the multicollinearity between the independent variables, income diversification, leverage, and total assets is tested using the Variance Inflation Factor method. The Variance Inflation Factor is a statistical method that helps in establishing the variance of the regression coefficient, which is affected by the multicollinearity of the independent variables used in the regression model. Generally, if the value is above 10, then the multicollinearity is high enough to cause an error in the regression model. According to the multicollinearity test, as shown in Table 4.5 above, the values are lower than the threshold value of 10.

This is because the value of 3.50 is obtained when calculating the value of leverage, while the value of 3.47 is obtained when calculating the value of total assets. In addition, the value of 2.84 is obtained when the value of income diversification is calculated, with a mean value of 3.27 for all the variables. This shows that although multicollinearity exists, it is at a level that will not affect the results obtained in the regression. In addition, the results obtained in the multicollinearity test indicate that the value of tolerance, given by $1/VIF$, is 0.285 for leverage, 0.288 for total assets, and 0.352 for income diversification, which are higher than 0.1, indicating that multicollinearity is not a major problem in the study data. This is good news concerning the results obtained in the panel regression, as it shows that all the variables will be able to make an independent contribution to the explanation of the variation in bank performance, and there is no problem of multicollinearity with other variables. The absence of multicollinearity is good news concerning the reliability of the results obtained in the regression analysis, as it shows that the results obtained in the regression will be valid, and the relationship between income diversification, leverage, total assets, and bank performance will be correctly interpreted.

4.7 Panel Unit Root Test

Table 4.6: Panel Unit Root Test Results (Levin–Lin–Chu Test)

Variable	Level / First Difference	LLC Test Statistic	p-value	Decision	Stationarity
ROA	Level	-3.214	0.0007	Reject H_0	Stationary
ROE	Level	-2.845	0.0022	Reject H_0	Stationary

Before carrying out the panel regression, it is vital to establish the stationarity of the variables to ensure the accuracy of the findings obtained in the regression analysis. This is in the sense that the non-stationarity of the variables could lead to misleading findings in the relationship between the variables in the regression analysis. For this study, the Levin-Lin-Chu panel unit root test was used to establish the stationarity of the dependent variables, ROA and ROE. The Levin-Lin-Chu panel unit root test is used to establish the null hypothesis that the panel data has a unit root, meaning the panel data is non-stationary, against the alternative hypothesis that the panel data is stationary. The findings obtained in the study, as presented in Table 4.6, show that the dependent variables, ROA and ROE, are stationary at the level. This is in the sense that the findings obtained in the study show that the dependent variables are stationary, meaning the mean and variance of the dependent variables are fixed and time-invariant, such that the effects of the changes in the dependent variables are short-run effects. The findings obtained in the study also show that the dependent variables are indeed stationary, meaning the findings obtained in the study show that the use of the fixed effects model and the random effects model is appropriate in the study. The study also reduces the possibility of spurious results, which could result in the trending nature of the variables, by establishing the stationarity of the major financial ratios and operational metrics used in the study.

The study was able to establish that the use of the fixed effects model was appropriate in the study, since the findings obtained in the study indicate that the dependent variables are indeed stationary, which implies that the mean and variance of the variables are fixed. In addition, the study was able to establish that the use of the random effects model was appropriate in the study, since the findings obtained in the study indicate that the dependent variables are indeed stationary.

4.8 Panel Data Regression Analysis

In order to assess the impact of income diversification, leverage, and total assets on bank performance, regression analysis was conducted using the panel regression results, and the results are presented below. According to the panel regression results, the impact of income diversification, leverage, and total assets on the return on assets

and return on equity of banks in Sri Lanka is revealed. Panel regression results are presented below:

4.8.1 ROA Model

Table 4.7: Fixed Effects Regression Results (Dependent Variable: ROA)

Variable	Coefficient	Std. Error	t-value	p-value
Income Diversification	-1.703	1.357	-1.25	0.217
Leverage	-1.097	1.623	-0.68	0.503
Total Assets	-1.35E-07	1.09E-07	-1.23	0.225
Constant	2.439	1.426	1.71	0.095

Statistic	Value
R ² (within)	0.0926
F-statistic	1.26
Prob > F	0.3028
Observations	50

Table 4.8: Random Effects Regression Results (Dependent Variable: ROA)

Variable	Coefficient	Std. Error	z-value	p-value
Income Diversification	-0.999	0.967	-1.03	0.301
Leverage	1.195	1.085	1.10	0.271
Total Assets	-8.75E-08	8.25E-08	-1.06	0.289
Constant	0.251	0.841	0.30	0.765

Statistic	Value
Wald Chi ²	1.83
Prob > Chi ²	0.6089
Observations	50

Table 4.9: Hausman Test for ROA Model

Test	Chi-square	p-value	Decision
Hausman Test	3.63	0.1631	Random Effects Preferred

The fixed effects regression results on the return on assets, as presented in the following Table 4.8, revealed that the coefficient is negative, -1.703, although the impact of income diversification is found to be insignificant on return on assets. In addition, the regression results revealed that the coefficient is negative, -1.097, although the impact of leverage is found to be insignificant on return on assets. Furthermore, the regression results revealed that the coefficient is negative, -1.35E-07, although the impact of total assets is found to be insignificant on return on assets. In addition, the regression results revealed that the R-squared is low, 0.0926, and the F-statistic is low, 1.26, and the p-value is 0.3028, indicating low explanatory power of the variables included in the regression results, as the p-value is higher, indicating that the variation in return on assets is explained by factors other than those included in the regression results.

The results of the Random Effects model for ROA, as shown in Table 4.9, indicate that income diversification results in negative values, -0.999, for the coefficient, while leverage results in positive values, 1.195, for the coefficient, with the coefficient for total assets being negative, -8.75E-08. However, none of the variables are significant at the 5% level of significance. Moreover, the Wald Chi² value is 1.83, with the p-value being 0.6089.

Table 4.10: Selected Model Summary (Random Effects – ROA)

Variable	Coefficient	Std. Error	p-value	Relationship
----------	-------------	------------	---------	--------------

Income Diversification	-0.999	0.967	0.301	Negative
Leverage	1.195	1.085	0.271	Positive
Total Assets	-8.75E-08	8.25E-08	0.289	Negative

Statistic	Value
Wald Chi ²	1.83
Prob > Chi ²	0.6089
Observations	50

The results of the Hausman test, as shown in Table 4.11, indicate that the Chi-square value for the test is 3.63, while the p-value is 0.1631, which indicates that the null hypothesis for the appropriateness of the Random Effects model cannot be rejected, as the p-value is greater than 0.05. Therefore, the Random Effects model is appropriate for the results, and the summary of the selected model, as shown in Table 4.11, indicates the negative relationship between income diversification, leverage, and the results for ROA, while the results also indicate that the asset utilization efficiency of the banks in Sri Lanka is not affected by the diversification strategies. Moreover, the results indicate that leveraging results in positive values for asset profitability.

4.8.2 ROE Model

Table 4.12: Fixed Effects Regression Results (Dependent Variable: ROE)

Variable	Coefficient	Std. Error	t-value	p-value
Income Diversification	-10.511	14.214	-0.74	0.464
Leverage	-7.316	16.992	-0.43	0.669
Total Assets	-1.05E-06	1.15E-06	-0.91	0.366

Constant	20.947	14.931	1.40	0.169
----------	--------	--------	------	-------

Statistic	Value
R ² (within)	0.0430
F-statistic	0.55
Prob > F	0.6487
Observations	50

Table 4.13: Random Effects Regression Results (Dependent Variable: ROE)

Variable	Coefficient	Std. Error	z-value	p-value
Income Diversification	-7.278	9.308	-0.78	0.434
Leverage	18.929	10.417	1.82	0.069
Total Assets	1.68E-07	8.11E-07	0.21	0.836
Constant	-4.213	7.689	-0.55	0.584

Statistic	Value
Wald Chi ²	6.05
Prob > Chi ²	0.1093
Observations	50

Table 4.14: Hausman Test for ROE Model

Test	Chi-square	p-value	Decision
------	------------	---------	----------

Hausman Test	3.92	0.1410	Random Effects Preferred
--------------	------	--------	--------------------------

Table 4.15: Selected Model Summary (Random Effects – ROE)

Variable	Coefficient	Std. Error	p-value	Relationship
Income Diversification	-7.278	9.308	0.434	Negative
Leverage	18.929	10.417	0.069	Positive
Total Assets	1.68E-07	8.11E-07	0.836	Positive

Statistic	Value
Wald Chi ²	6.05
Prob > Chi ²	0.1093
Observations	50

The results for the Fixed Effects model for ROE, as shown in Table 4.12, indicate that income diversification, leverage, and the asset utilization efficiency of the banks in Sri Lanka have negative values for the coefficient, while none of the variables are significant at the 5% level of significance. Moreover, the results indicate that the within R² value is low, at 0.043, while the results also indicate that the F-statistic value is low, at 0.55, with the p-value being 0.6487, which indicates that the variations in the results for ROE for the sample banks cannot be explained by the variables included in the regression results.

On the other hand, the results of the Random Effects regression model for ROE, as presented in Table 4.13, revealed that the relationship between leverage (18.929) and ROE is slightly significant at ($p = 0.069$), indicating that banks with higher debt levels tend to have higher rates of returns to their shareholders. On the other hand, the effect of income diversification is still nonsignificant and negative (-7.278), and the effect of

total assets ($1.68E-07$) is positive and negligible. The Wald Chi² test is significant at 6.05 ($p = 0.1093$).

The results of the Hausman test, as presented in Table 4.14, revealed that the Chi-square statistic is 3.92, and the p-value is 0.1410, indicating that it is higher than 0.05, and as such, the null hypothesis cannot be rejected, indicating that the Random Effects regression model is appropriate for the study of ROE, whereas the results of the selected model, as presented in Table 4.15, revealed that income diversification is nonsignificant and negative, whereas leverage is marginal and positive, indicating that diversification strategies have no impact on ROE, whereas leverage has a positive impact on ROE.

The results obtained in the empirical model of the effect of income diversification on the performance of banks in Sri Lanka have given rise to several important findings. From the results obtained in the estimation of the models of ROA and ROE using the Fixed and Random Effects methods, it is evident that there is a negative correlation between the income diversification and the performance of the bank, although the results obtained were statistically insignificant. This is similar to the results obtained in the results of the study, indicating that banks in emerging markets that obtain a higher proportion of their total income from sources other than interest are not likely to obtain enhanced bank and shareholder efficiency in terms of profitability. This is similar to the results obtained in the results of the study, indicating that diversification of the bank's income to non-traditional sources, such as fees and commissions, is likely to have a negative effect on the financial performance of the bank due to the higher operating costs of the bank. However, the negative sign also indicates that there is a likely problem in the Sri Lankan banking sector in managing the effect of income diversification, where the traditional banking activities are the source of the bank's profitability.

Leverage is also revealed as an important variable in explaining bank performance, particularly in the case of the ROE model. Although the coefficient of leverage is not significant in the case of the ROA model, it is marginally significant in the case of the ROE model, indicating that banks with higher leverage tend to achieve higher returns to their shareholders. This is in line with the theoretical expectation that the application of leverage results in higher returns to the shareholders of the firm, despite the higher risk. However, the non-significant coefficient of leverage in the case of the ROA model indicates that the application of leverage does not enhance the efficiency of asset

utilization, although it enhances the distribution of profits to the shareholders of the firm. Total assets, used as an alternative variable to bank size, is revealed to have a negligible effect on the determination of bank performance in terms of both ROA and ROE, indicating that bank size does not have an inherent effect on bank performance. This is in line with the theoretical expectation that bank size does not enhance bank performance, particularly bank profitability, after a certain level is attained.

The results obtained in the correlation analysis are also in line with the findings obtained in the regression results. In the correlation results, it is revealed that there is a strong positive correlation between the bank performance variables, particularly between ROA and ROE, indicating that bank performance is enhanced with the level of asset utilization efficiency. On the other hand, the results obtained in the correlation results indicate that there is a very low correlation between income diversification and bank profitability, indicating that income diversification does not have a significant relationship with bank profitability. Although the results obtained in the multicollinearity results indicate that the values of all the variables are below the critical value, the results obtained in the panel unit root test indicate that the bank performance variables, particularly ROA and ROE, are stationary at level, indicating that the regression results could be used without the risk of spurious relationships.

4.10 Summery of findings

As per the panel regression results, the following key findings can be derived. Firstly, income diversification is not statistically significant for ROA and ROE. This suggests that banks that earn their income through income other than interest, such as fees, commissions, etc., may not necessarily earn high profitability, which reiterates the importance of traditional lending activities for banks. Secondly, financial leverage is related to ROE, which is statistically significant, suggesting that banks can increase their shareholder value through financial leverage. However, financial leverage is not related to ROA. Thirdly, bank size, as measured through total assets, does not have much impact on ROA and ROE, suggesting that banks that are larger in size may not necessarily have high profitability. Fourthly, the Random Effects model is better than the Fixed Effects model for explaining ROA and ROE, as suggested by the results of

the statistical tests, such as the Hausman test, which suggests that the variables that have been omitted from the model have no correlation with the independent variables.

4.11 Chapter Summary

This chapter gave an exhaustive analysis of the impact of income diversification, leverage, and bank size on the performance of banks in Sri Lanka for the period of 2019 to 2023. Descriptive statistical results indicated the general financial characteristics of the selected banks, with an average ROA of 0.97%, an average ROE of 11.08%, and a mean income diversification ratio of 0.187. Normality tests were conducted and revealed that the variables were appropriate for regression analysis, and multicollinearity tests revealed that the variables did not have any major correlation with each other. Panel unit root tests also revealed the stationarity of the dependent variables, making the results more reliable for the purpose of this study. The Fixed Effects and Random Effects models were applied to the variables of bank performance, namely ROA and ROE. The results revealed that, according to the results of the Hausman tests, the Random Effects models were more appropriate for the purpose of this study, as it included both cross-sectional and interbank variations. The results revealed that income diversification had an insignificant negative impact on bank performance, whereas leverage had an insignificant positive impact on bank performance, and bank size had the least impact on bank performance. Banks in Sri Lanka still depend on their traditional interest-based business for better performance, whereas income diversification, though theoretically sound, does not impact bank performance in any way, as revealed in the results of the study. The insights revealed in this chapter form the basis for the conclusions and recommendations revealed in Chapter Five, focusing on the implications of income diversification and capital structure for strategic management in the banking industry.

CHAPTER FIVE

CONCLUSIONS AND SUGGESTIONS

5.1 Chapter Introduction

This chapter is based on the conclusions, implications, and recommendations derived from the study based on the findings. The study aims to investigate the impact of income diversification on bank performance for licensed commercial banks in Sri Lanka for the years 2019-2023. In this study, the researcher used panel data collected from ten banks, including state and private domestic banks. In addition, the study used the Fixed Effects (FE) model and the Random Effects (RE) model to investigate the relationship between income diversification, leverage, bank size, and bank performance, which is measured by return on assets and return on equity. This chapter is based on the findings derived from the descriptive statistics, correlation analysis, multicollinearity tests, panel unit root tests, panel regression models, which provide valuable insights about the study on income diversification. This chapter is based on the conclusions derived from the study, which provide valuable implications for bank management. Moreover, the conclusions derived from this chapter are based on the study's objectives and research questions, which provide an overall understanding about the study on the impact of income diversification, along with the control variables leverage and total assets, on commercial banks in Sri Lanka.

5.2 Conclusions

From the results generated from the analysis of the panel data for the five-year period, several important insights into the relationship between income diversification and bank performance in Sri Lanka have been generated. Firstly, the income diversification, as indicated by the ratio of non-interest income to total operating income, was revealed to have had a negative relationship with both ROA and ROE, although the results were found to be statistically insignificant in all the regression models. This implies that the efforts put in place by the banks to diversify their income sources beyond the traditional

interest-based income-generating activities have been unsuccessful in improving the asset efficiency and shareholder returns of the banks. This is in conformity with the findings of the results generated from the studies conducted in the emerging economies, which also suggested that income diversification may not have a positive impact on the profitability of the banks, especially when the traditional lending activities of the banks dominate their income-generating activities.

The results generated from the analysis revealed that the variable leverage had a positive relationship with ROE although the results were found to be insignificant when the variable was subjected to the regression model with ROA. This implies that the banks that adopted high levels of leverage in their capital base were able to achieve high levels of shareholder returns through the efficient utilization of the borrowed funds in their lending and investment activities. The results also suggest that the variable leverage had no significant impact on asset efficiency, suggesting that the banks were at risk due to the excessive adoption of debt finance in their capital base.

The results generated from the analysis revealed that the variable total assets, representing the overall size of the banks, had negligible effects on both ROA and ROE. This implies that the larger banks were unable to achieve high levels of profitability relative to the smaller banks, suggesting that the overall size of the bank has no significant influence on the overall performance of the bank, although this is highly debatable.

Moreover, the results of the Hausman specification tests revealed that the Random Effects Model is appropriate for performing ROA and ROE regression analyses. This implies that, apart from other factors, differences in bank performance are largely driven by decisions made by banks rather than inherent characteristics of banks. Therefore, in conclusion, it is evident that the impact of income diversification on bank performance is insignificant, and banks should diversify their income with caution, as it should be balanced with their banking operations and other prudent banking practices.

5.3 Suggestions

Several suggestions can be made based on the findings, and they may be applicable to bank managers, policymakers, and regulators. First, it is evident that income

diversification does not impact bank performance, and banks should diversify their income as it is an important aspect that complements their traditional banking operations. Banks should diversify their income into other income-generating activities such as fee income, commission income, and treasury income, as it is expected to positively impact their profitability, particularly when the rates of interest are subject to fluctuations. However, banks should be cautious when diversifying their income so that it does not compromise their traditional banking operations.

Banks should be careful about their level of leverage, which has a slight positive impact on shareholder returns. Banks should be aware that, though higher levels of leverage have positive benefits, they should be careful about the risks that come with it, such as liquidity risk and credit risk. The Central Bank of Sri Lanka, which regulates banks, should be aware of this issue and provide guidelines to banks on how to manage their capital structure so that they can be profitable.

Banks should be aware that they can achieve efficiency instead of depending on their size to achieve performance. Banks can achieve efficiency through investments in technology, which will eventually lead to diversification and an increase in profitability ratios of banks, such as ROA and ROE. Banks can also be aware of their financial ratios, which will eventually lead to achieving performance through identification of areas that need attention.

The government should be aware of this issue and should promote diversification through the introduction of policies that will eventually encourage banks to adopt new financial products. For instance, the government should introduce policies that will promote microfinance, SME loans, and other financial products introduced through technology. Banks will be able to diversify their income without compromising their profits. The income will be managed through financial literacy.

The scope of the research can also be expanded to incorporate other variables, such as the role played by foreign banks operating in Sri Lanka. The scholars can also incorporate other control variables, which will eventually help in attaining better results. For example, the research can also be conducted on the role played by liquidity risk, ownership, etc., which will eventually influence the performance of the banks. The scholars can also use other econometric models, which will eventually help in attaining

better results. For example, the scholars can also use the GMM model to attain better results.

5.4 Chapter Summary

It is evident that income diversification has a limited and insignificant impact on the performance of banks in Sri Lanka. Leverage has a slightly positive impact on ROE, while bank size is not significant in determining profitability. This study is significant and valuable to bank managers, regulators, and policymakers, as it offers valuable insights into the ways in which strategies can be formulated to enhance the performance of banks, maintaining stability in the banking sector. This chapter has emphasized the need to conduct more research with more variables, time intervals, and sophisticated models to understand the impact of income diversification on the performance of banks.

References

- Jayasena, D. R. (2018). Impact of income diversification on bank performance: special references to licensed commercial banks in Sri Lanka.
- Kurincheedaran, S. (n.d.). Sectoral diversification and bank performance: An empirical study on domestic licensed commercial banks in Sri Lanka. *International Journal of Accounting & Business Finance*, 2015.
- Menike, M. G. (2024). The impact of portfolio diversification on the profitability of licensed commercial banks in Sri Lanka. *Journal of Accounting and Finance*, 11(2), 110-120.
- Cuong Van Hoang, L. Q. T. N., Tran, M. D., & Hoang, T. D. (2020). The impact of income diversification on liquidity creation and financial performance of Vietnamese commercial banks. *International Journal of Financial Research*, 11(2), 1–15.
- Jayasena, D. R. P. L., & Amarasinghe, M. A. A. D. (2018, October 23). Impact of income diversification on bank performance: Special references to licensed commercial banks in Sri Lanka. 3rd Interdisciplinary Conference of Management Researchers, Sabaragamuwa University of Sri Lanka. <http://repo.lib.sab.ac.lk:8080/xmlui/handle/123456789/115>
- Kurincheedaran, S. (2015, December 20). Sectoral diversification and bank performance: An empirical study on domestic licensed commercial banks in Sri Lanka. *International Journal of Accounting & Business Finance*, 1(2). <https://ssrn.com/abstract=2712098>
- Menike, M. G. P. D. (2022). The impact of portfolio diversification on the profitability of licensed commercial banks in Sri Lanka. University of Peradeniya.
- Pisedtasalasai, A., & Edirisuriya, P. (2020). Diversification and performance of Sri Lankan banks. *Journal of Asian Finance, Economics and Business*, 7(9), 1–10. <https://doi.org/10.13106/JAFEB.2020.VOL7.NO9.001>

- Jayasena, D. R. (2018). Impact of income diversification on bank performance: special references to licensed commercial banks in Sri Lanka.
- Kurincheedaran, S. (n.d.). Sectoral diversification and bank performance: An empirical study on domestic licensed commercial banks in Sri Lanka. *International Journal of Accounting & Business Finance*, 2015.
- Menike, M. G. (2024). The impact of portfolio diversification on the profitability of licensed commercial banks in Sri Lanka. *Journal of Accounting and Finance*, 11(2), 110-120.
- Cuong Van Hoang, L. Q. T. N., Tran, M. D., & Hoang, T. D. (2020). The impact of income diversification on liquidity creation and financial performance of Vietnamese commercial banks. *International Journal of Financial Research*, 11(2), 1–15.
- Jayasena, D. R. P. L., & Amarasinghe, M. A. A. D. (2018, October 23). Impact of income diversification on bank performance: Special references to licensed commercial banks in Sri Lanka. 3rd Interdisciplinary Conference of Management Researchers, Sabaragamuwa University of Sri Lanka. <http://repo.lib.sab.ac.lk:8080/xmlui/handle/123456789/115>
- Kurincheedaran, S. (2015, December 20). Sectoral diversification and bank performance: An empirical study on domestic licensed commercial banks in Sri Lanka. *International Journal of Accounting & Business Finance*, 1(2). <https://ssrn.com/abstract=2712098>
- Menike, M. G. P. D. (2022). The impact of portfolio diversification on the profitability of licensed commercial banks in Sri Lanka. University of Peradeniya.
- Pisedtasalasai, A., & Edirisuriya, P. (2020). Diversification and performance of Sri Lankan banks. *Journal of Asian Finance, Economics and Business*, 7(9), 1–10. <https://doi.org/10.13106/JAFEB.2020.VOL7.NO9.001>
- Athanasoglou, P. P., Brissimis, S. N., & Delis, M. D. (2008). Bank-specific, industry-specific and macroeconomic determinants of bank profitability. *Journal of International Financial Markets, Institutions and Money*, 18(2), 121–136. <https://doi.org/10.1016/j.intfin.2006.07.001>

- Berger, A. N., & Bouwman, C. H. S. (2009). Bank liquidity creation. *Review of Financial Studies*, 22(9), 3779–3837. <https://doi.org/10.1093/rfs/hhn104>
- Berger, A. N., Hasan, I., & Zhou, M. (2010). The effects of focus versus diversification on bank performance: Evidence from Chinese banks. *Journal of Banking & Finance*, 34(7), 1417–1435. <https://doi.org/10.1016/j.jbankfin.2010.01.010>
- Berger, A. N., Humphrey, D. B., & Pulley, L. B. (1999). Do consumers pay for one-stop banking? *Journal of Banking & Finance*, 23(2–4), 275–301. [https://doi.org/10.1016/S0378-4266\(98\)00071-4](https://doi.org/10.1016/S0378-4266(98)00071-4)
- Bliss, R. T., & Rosen, R. J. (2001). CEO compensation and bank mergers. *Journal of Financial Economics*, 61(1), 107–138. [https://doi.org/10.1016/S0304-405X\(01\)00061-5](https://doi.org/10.1016/S0304-405X(01)00061-5)
- Bryant, J. (1980). A model of reserves, bank runs, and deposit insurance. *Journal of Banking & Finance*, 4(4), 335–344. [https://doi.org/10.1016/0378-4266\(80\)90012-6](https://doi.org/10.1016/0378-4266(80)90012-6)
- Diamond, D. W., & Dybvig, P. H. (1983). Bank runs, deposit insurance, and liquidity. *Journal of Political Economy*, 91(3), 401–419. <https://www.jstor.org/stable/1837095>
- DeYoung, R., & Roland, K. P. (2001). Product mix and earnings volatility at commercial banks. *Journal of Financial Intermediation*, 10(1), 54–84. <https://doi.org/10.1006/jfin.2000.0305>
- Edirisuriya, P., Gunasekarage, A., & Perera, S. (2019). Product diversification and bank risk: Evidence from South Asian banking institutions. *Applied Economics*, 51(15), 1626–1643. <https://doi.org/10.1080/00036846.2018.1528341>
- Ekanayake, W., & Wanniarachchige, M. K. (2023). Effect of ownership on the performance of licensed commercial banks in Sri Lanka. <http://repository.rjt.ac.lk/handle/123456789/1730>

- Goddard, J., Molyneux, P., & Wilson, J. O. S. (2004). Dynamics of growth and profitability in banking. *Journal of Money, Credit and Banking*, 36(6), 1069–1090. <https://www.jstor.org/stable/3839025>
- Hoanga, C. V., Nguyena, L. Q. T., Trana, M. D., & Hoanga, T. D. (2020). The impact of income diversification on liquidity creation and financial performance of Vietnamese commercial banks. *Accounting*, 6(3), 347–356. <https://doi.org/10.5267/j.ac.2020.4.004> (PDF) <https://pdfs.semanticscholar.org/>
- IMF. (2017). *Global Financial Stability Report*. International Monetary Fund. <https://www.imf.org/en/Publications/GFSR>